

Artificial Intelligence

Lecture 1: Introduction

Xiaojin Gong

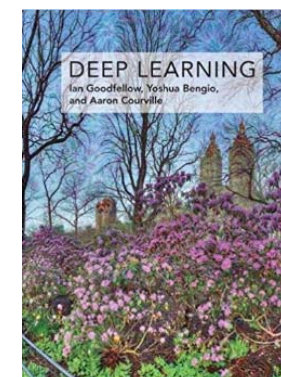
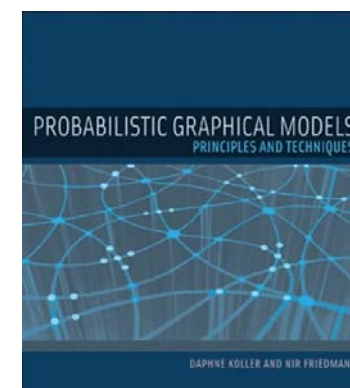
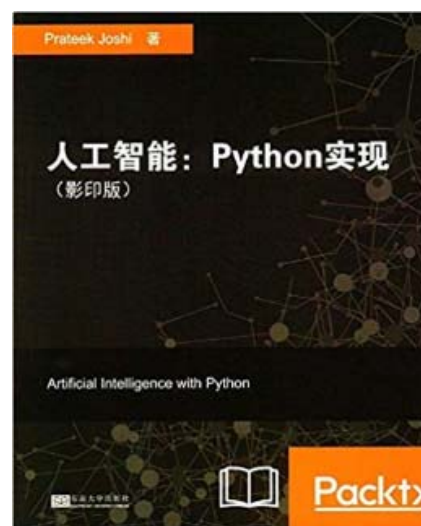
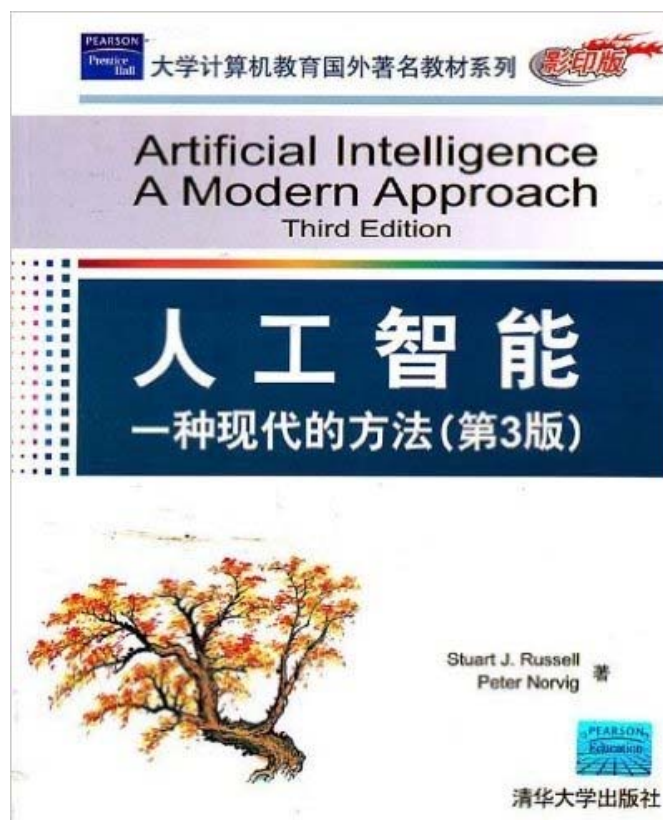
2021-03-01

Outline

- Course Overview
- What is AI?
- AI History
- Course Topics
- Intelligent Agents

Course Overview

- Course Website:
 - 学在浙大 <http://course.zju.edu.cn/>
- Textbook: <http://aima.cs.berkeley.edu/index.html>



Course Overview

■ 2017级信息工程专业培养方案

(3) 专业模块课程 16.5 学分↵

1) 课程组一 课程组一 ≥6 学分↵

A. 在以下课程中至少修读 1 门 3 学分↵

课程号	课程名称	学分	周学时	年级	学期↵
67190050	数据分析与算法设计	3.0	3.0-0.0	三	秋冬↵
67190130	射频电路与系统	3.0	3.0-0.0	三	秋冬↵
67190060	网络基础	3.0	3.0-0.0	三	春夏↵

B. 其他课程↵

课程号	课程名称	学分	周学时	年级	学期↵
67190020	计算机组成与设计	3.5	3.0-1.0	三	秋冬↵
67190030	数字图像处理	3.0	3.0-0.0	三	春夏↵
67190040	人工智能	3.0	3.0-0.0	三	春夏↵

Course Overview

■ Undergraduate Artificial Intelligence Program at CMU

Math and Statistics

All of the following:

15-151	Mathematical Foundations for Computer Science (if not offered, substitute 21-127)
21-120	Differential and Integral Calculus
21-122	Integration and Approximation
21-241	Matrices and Linear Transformations
36-218	Probability Theory for Computer Scientists or: (15-259 or 21-325 or 36-225) and 36-226
36-401	Modern Regression

Computer Science

All of the following:

07-128	Freshman Immigration Course
15-122	Principles of Imperative Computation
15-150	Principles of Functional Programming
15-210	Parallel and Sequential Data Structures and Algorithms
15-213	Introduction to Computer Systems
15-251	Great Ideas in Theoretical Computer Science

Artificial Intelligence

All of the following three AI core courses:

07-180	Concepts in Artificial Intelligence
10-401	Introduction to Machine Learning (Undergrad)
15-381	Artificial Intelligence: Representation and Problem Solving

plus one of the following AI core courses:

11-411	Natural Language Processing
16-385	Computer Vision

What is AI?



全球首个获得公民身份的机器人
2017年 沙特索菲亚



国产骨科手术机器人“天玑”
2018年

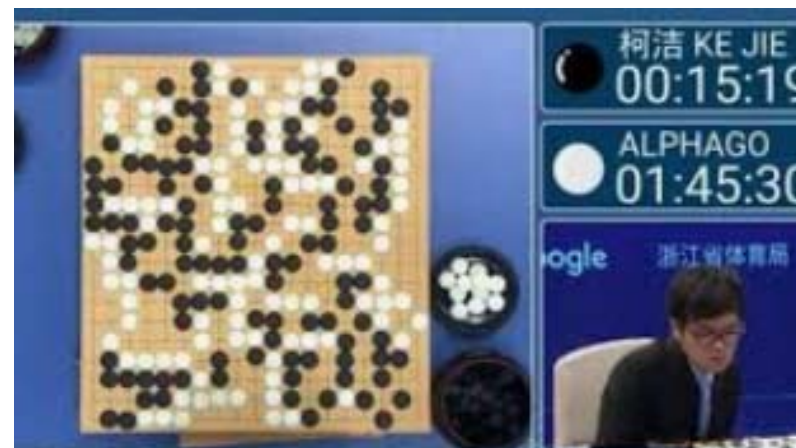


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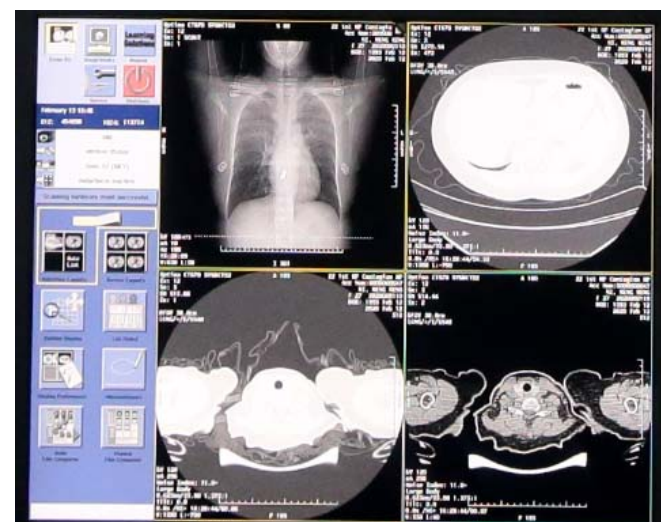


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What is AI?



中科天玑 “智疫通”



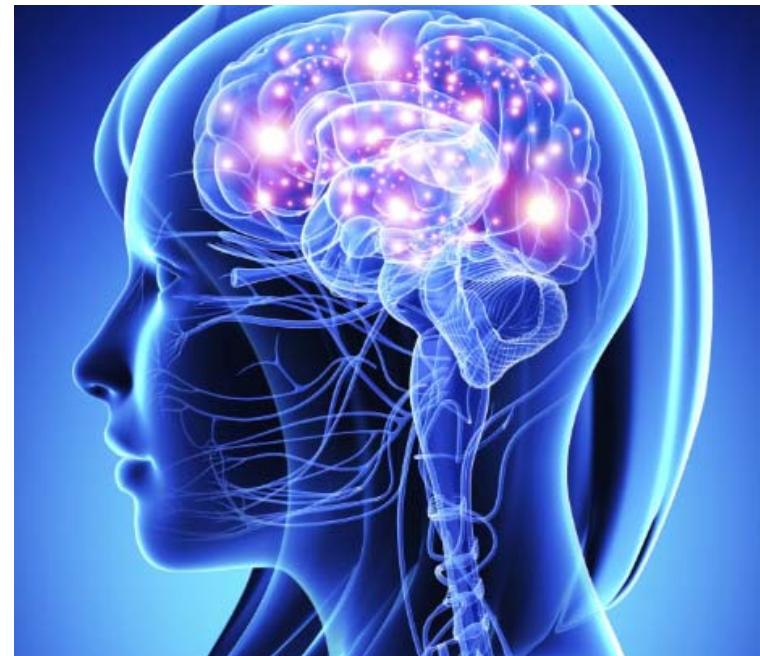
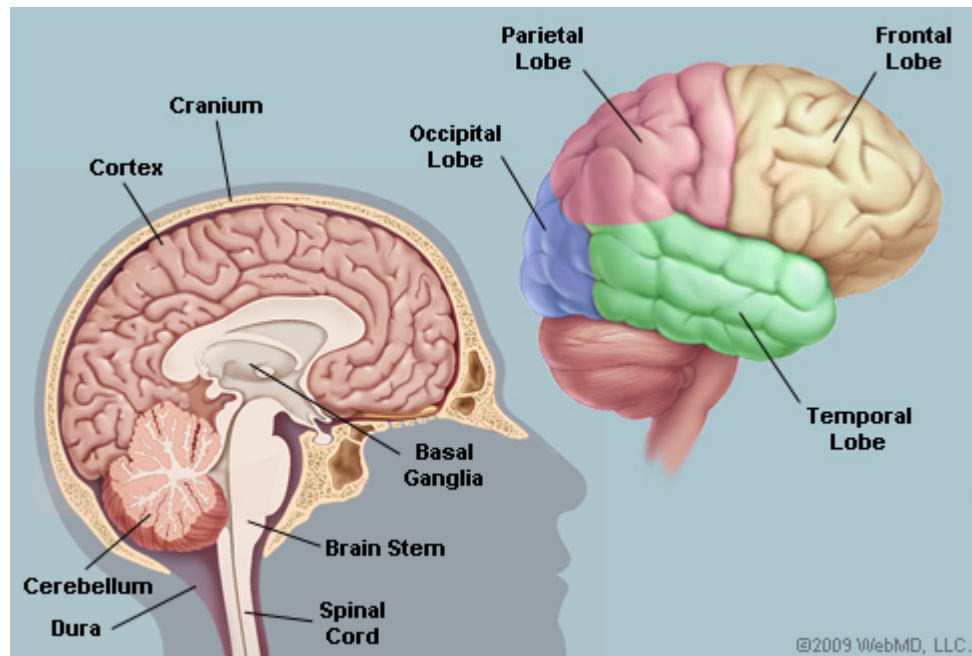
阿里达摩院AI诊断系统

What is AI?

Thinking Humanly “The exciting new effort to make computers think ... <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI ... is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

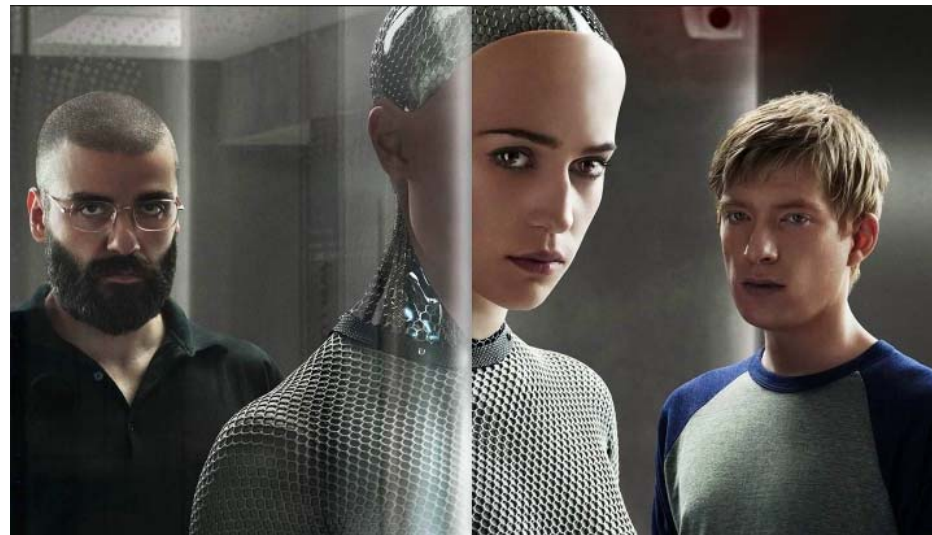
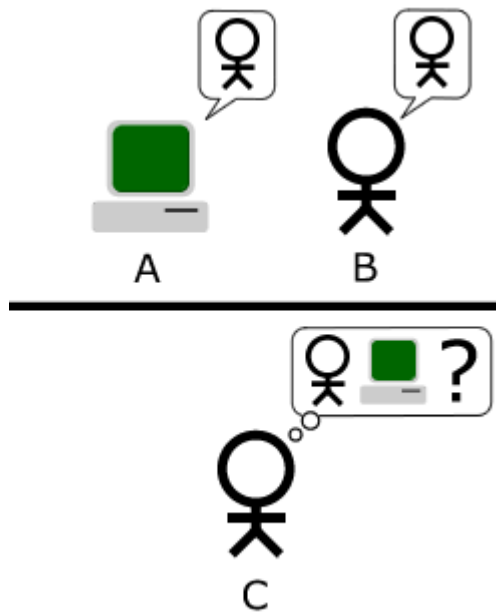
Thinking Humanly

- Cognitive science
 - Introspection
 - Psychological experiments
 - Brain imaging



Acting Humanly

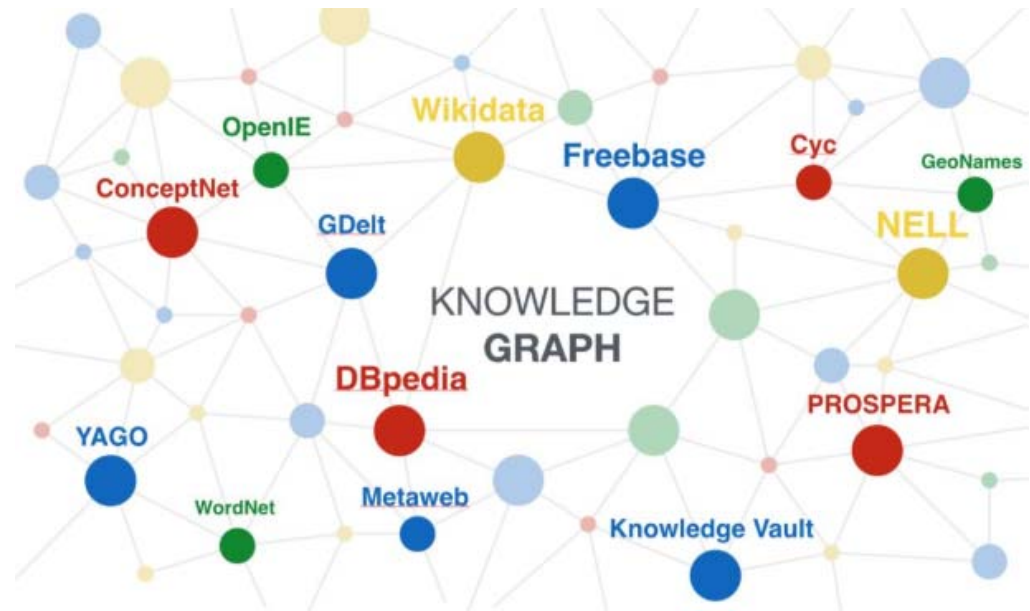
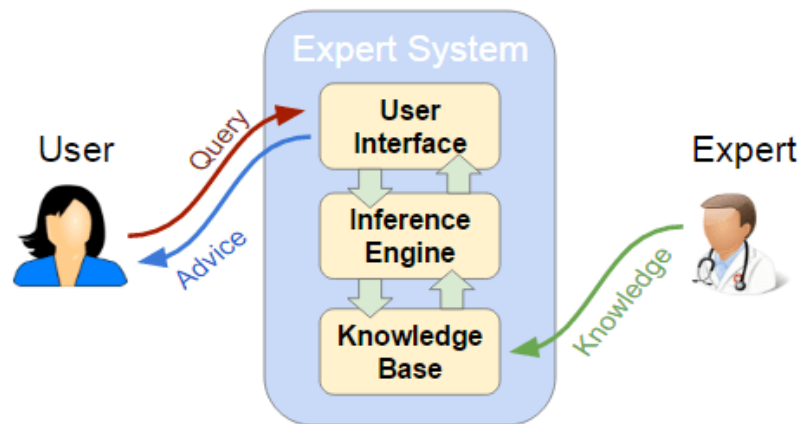
- Turing test: (Alan Turing, 1950)
 - Natural language processing
 - Knowledge representation
 - Automated reasoning
 - Machine learning
 - Computer vision
 - Robotics



Ex Machina (2015)

Thinking Rationally

- Logic
 - Aristotle: Syllogisms



Acting Rationally

- Rational agent
 - Acts so as to achieve the best outcome, or when there is uncertainty, the best expected outcome.

This course concentrates on general principles of rational agents and on components for constructing them.

Philosophical Foundations of AI?

- Strong AI
 - Can machines really think?
 - Conscious machines, self-awareness
 - Little progress, little serious activity, controversial
- Weak AI
 - Can machines act intelligently?
 - Focused on specific tasks
 - Get machines to do things that people can do which machines currently cannot
 - Most AI research on this topic
 - Lots of progress recently

Philosophical Foundations of AI?

- AI Safety, Ethics, and Policy

NEW

Just In

Print

Bill Gates joins Stephen Hawking in warning Artificial Intelligence IS a threat to mankind

BILL Gates has warned that Artificial Intelligence (AI) is a REAL threat to mankind.

By **DION DASSANAYAKE**
PUBLISHED: 22:09, Thu, Jan 29, 2015

SHARE **TWEET** 291

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Updated 3 Dec 2

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GETTY / UNIVERSAL

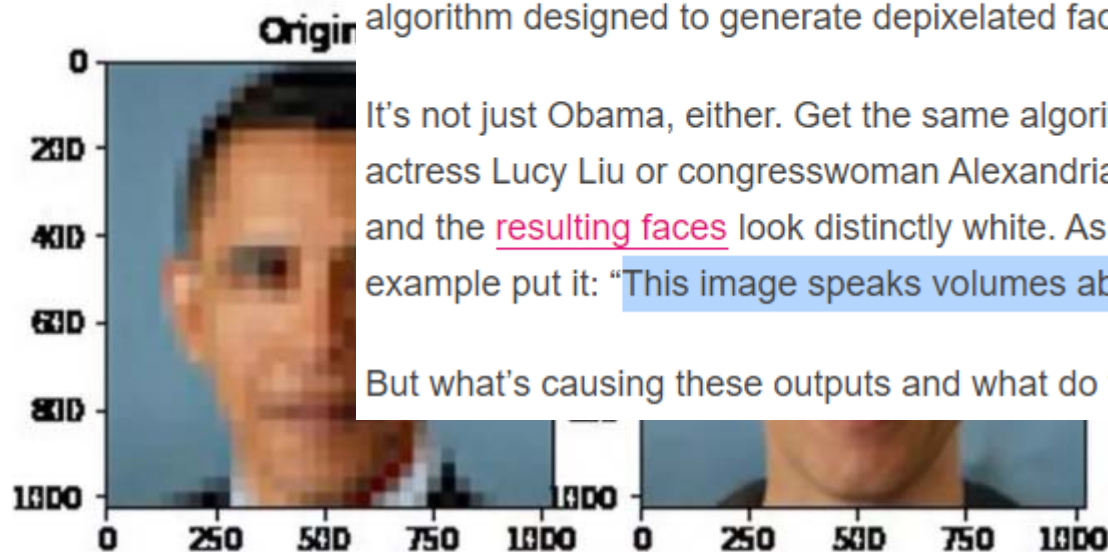
Bill Gates and Alicia Vikander in Ex Machina, a film that looks at how AI affects humans

as concerns

Philosophical Foundations of AI?

- AI Safety, Ethics, and Policy
 - Fairness, bias, inequality
 - Privacy and National Security
 - Rights and moral consideration for AI and robots?
 - Ethics and AI: teaching machines to be moral?

It's a startling image that illustrates the deep-rooted biases of AI research. Input a low-resolution picture of Barack Obama, the first black president of the United States, into an algorithm designed to generate depixelated faces, and the output is a white man.



It's not just Obama, either. Get the same algorithm to generate high-resolution images of actress Lucy Liu or congresswoman Alexandria Ocasio-Cortez from low-resolution inputs, and the resulting faces look distinctly white. As one popular tweet quoting the Obama example put it: "This image speaks volumes about the dangers of bias in AI."

But what's causing these outputs and what do they really tell us about AI bias?

Philosophical Foundations of AI?

- AI Safety, Ethics, and [
- Fairness, bias, inequality
- Privacy and National S
- Rights and moral consi
- Ethics and AI: teaching



杭州市智能网联汽车开放测试道路公示

根据杭州市经济和信息化局、杭州市公安局、杭州市交通运输局《关于印发杭州市智能网联车辆道路测试管理实施细则（试行）的通知》（杭经信联推进〔2018〕121号）要求，经第三方测试机构提供道路环境勘察报告，专家评估，杭州市自动驾驶测试管理联席工作小组研究，现就杭州市余杭区未来科技城文二西路及周边道路开放用于智能网联汽车测试路段进行公示。

一、道路范围

1. 文二西路（西：绿汀路，东：高教路）
2. 葛巷路（西：绿汀路，东：景兴路）
3. 绿汀路（北：文一西路，南：文二西路）
4. 创明路（北：葛巷路，南：文二西路）
5. 景兴路（北：葛巷路，南：文二西路）

二、测试时间

许可测试时间为 10:00-16:00、21:00-00:00。且无天气预报等级为暴雨、暴雪、能见度低于 100 米的大雾等极端天气情况。

公示期为：2019 年 5 月 28 日到 2019 年 6 月 3 日。

杭州市自动驾驶测试管理联席工作小组



2019 年 5 月 28 日

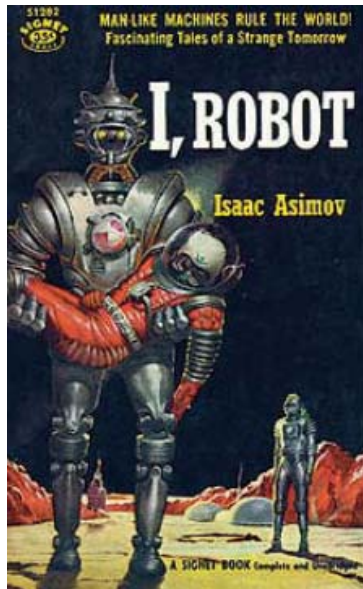
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review

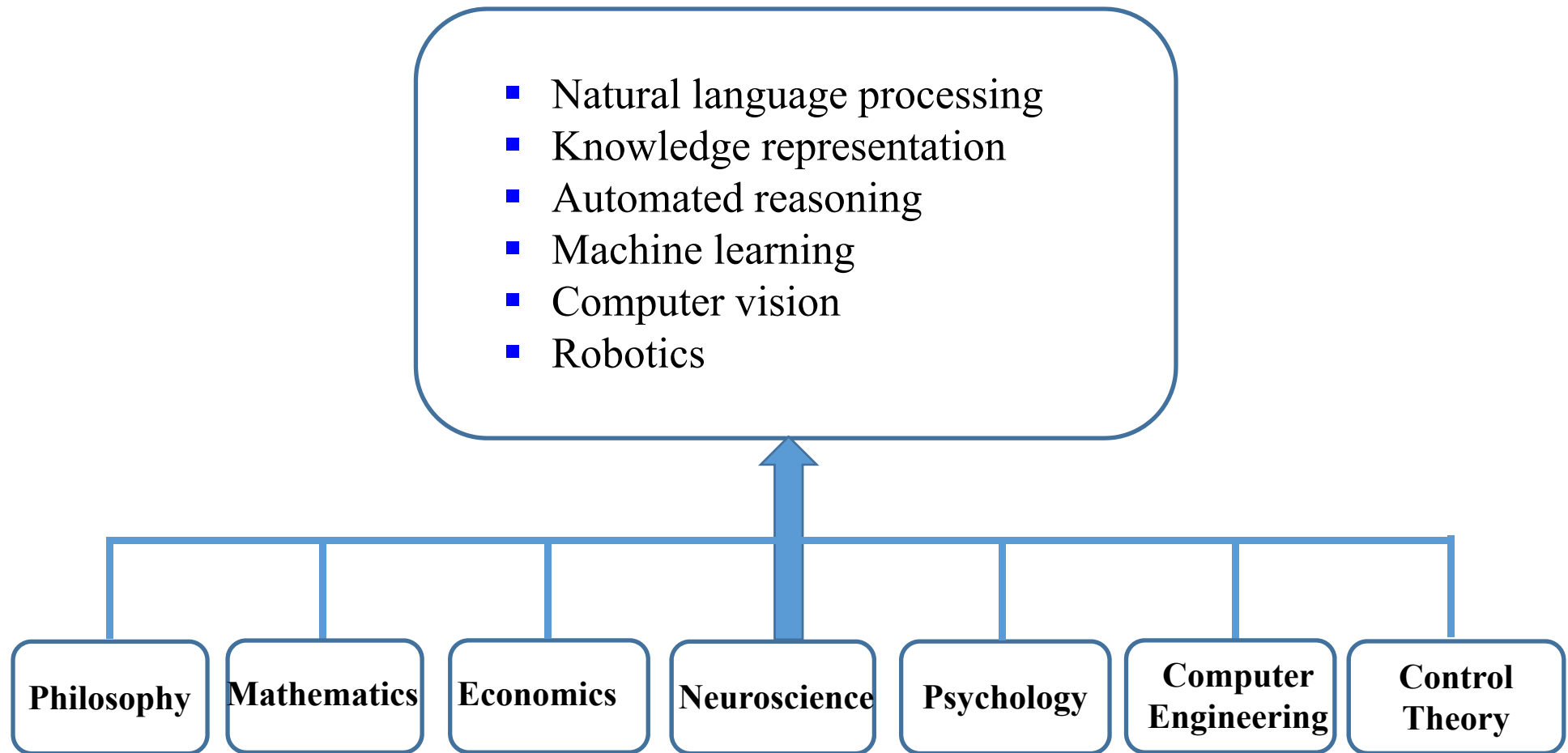
hat will bring lifesaving
ed to develop new solution
tentia benefits for safety,

Philosophical Foundations of AI?

- AI Safety, Ethics, and Policy
 - Isaac Asimov's "Three Laws of Robotics"
 - A robot may not injure a human being or, through inaction, allow a human being to come to harm.
 - A robot must obey orders given it by human beings except where such orders would conflict with the First Law.
 - A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

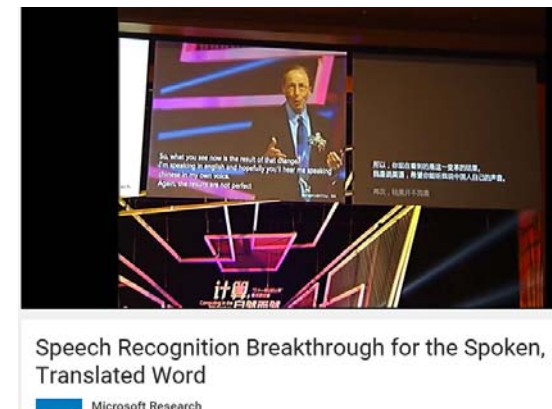
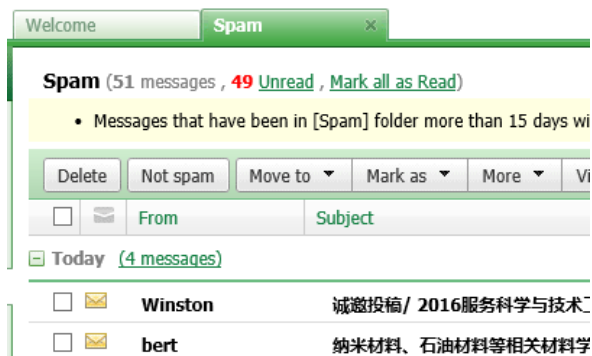


The Foundations of AI



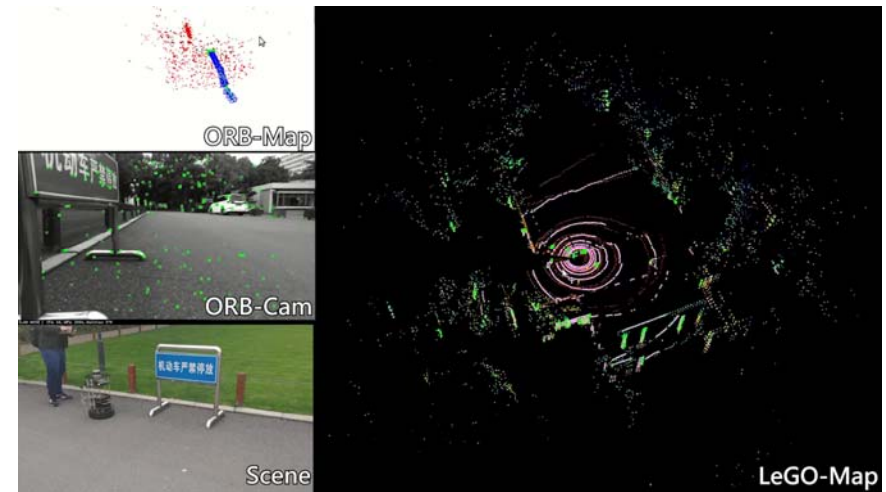
Natural Language Processing

- Automatic speech recognition
- Text-to-speech synthesis
- Dialog systems
- Machine translation
- Spam filtering
- ...



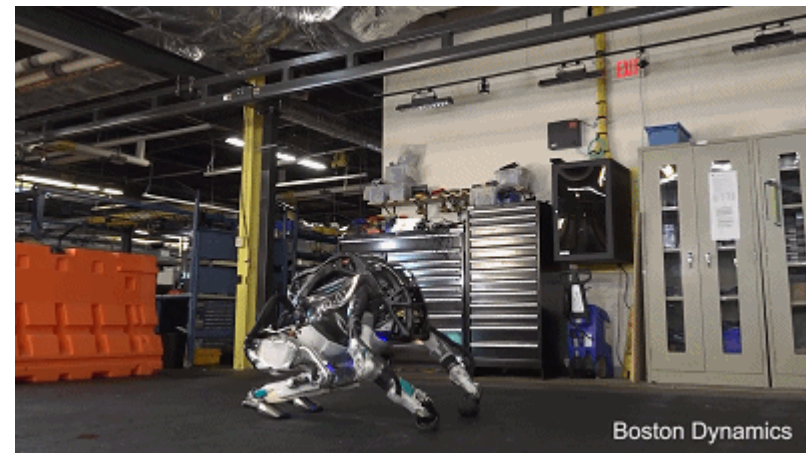
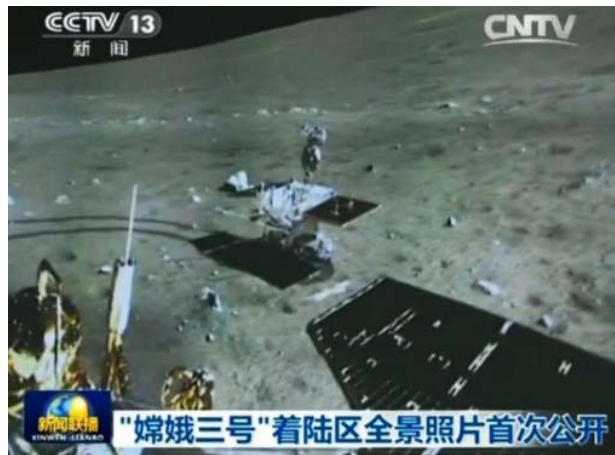
Computer Vision

- Object detection/recognition
- Scene understanding
- Image classification
- Pose recognition
- ...

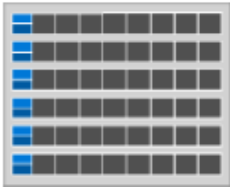
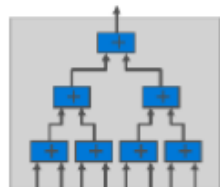
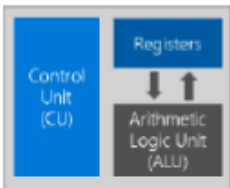
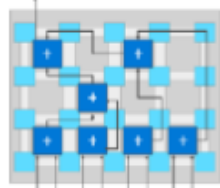


Robotics

- Autonomous driving
- Unmanned ground vehicles
- Autonomous underwater vehicles
- ...
 - Environment perception
 - Map building
 - Path planning



AI Chips

	训练端	推理端	
	GPU ：以英伟达为主，AMD为辅标榜通用性，多维计算及大规模并行计算架构契合深度学习的需要。在深度学习上游训练端（主要用在云计算数据中心里），GPU是当仁不让的第一选择。	GPU ：英伟达Volta GPU也开始布局推理端。深度学习下游推理端虽可容纳CPU/FPGA/ASIC等芯片，但竞争态势中英伟达依然占主导。	
	ASIC ：以谷歌的TPU、英特尔的Nervana Engine为代表，针对特定框架进行深度优化定制。但开发周期较长，通用性较低。比特币挖矿目前使用ASIC专门定制化矿机。	ASIC ：下游推理端更接近终端应用，需求也更加细分，英伟达的DLA，寒武纪的NPU等逐步面市，将依靠特定优化和效能优势，未来在深度学习领域分一杯羹。	
	CPU ：通用性强，但难以适应于人工智能时代大数据并行计算工作。	FPGA ：依靠可编程性及电路级别的通用性，适用于开发周期较短的IoT产品、传感器数据预处理工作以及小型开发试错升级迭代阶段等。但较成熟的量产设备多采用ASIC。	



Credits: 天风证券研究所

AI Chips

Cambricon
寒 武 纪

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云端AI加速卡

为企业级数据中心提供AI计算加速

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智能处理器IP

采用定制化的低功耗处理器架构，广泛应用于各类智能终端

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边缘AI加速卡

面向超紧凑/低功耗的边缘端AI推理部署方案

[了解 更多](#)

The History of AI

A PROPOSAL FOR THE DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

J. McCarthy, Dartmouth College
M. L. Minsky, Harvard University
N. Rochester, I.B.M. Corporation
C.E. Shannon, Bell Telephone Laboratories

August 31, 1955

1956 Dartmouth Conference: The Founding Fathers of AI



John McCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff



Alan Newell



Herbert Simon



Arthur Samuel



Oliver Selfridge



Nathaniel Rochester



Trenchard More

The History of AI

1956

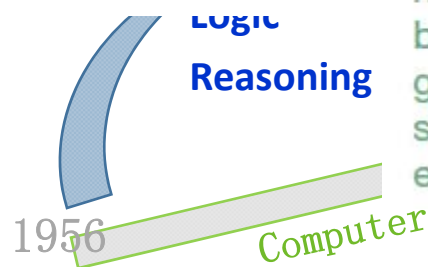
The term “artificial intelligence” is coined at a Dartmouth conference and AI is founded as an academic discipline.

1956–1974

The golden years of AI era, characterized by increased funding in promising, logic-based, problem-solving approaches.

1974–1980

Overly high expectations about the limited capacities of AI leads to the first “AI winter” with reduced funding and interest in AI.



1980–1987

The rise of knowledge-based expert systems brings new successes and a change in the focus of research and funding toward this form of AI.

1987–1993

The second “AI winter” starts with a sudden collapse of the special interest hardware industry in 1987. This brings with it negative perceptions of AI in governments and investors, as well as systems show their limitations and are expensive to update and maintain.

1993–2011

Optimism about AI returns and increases. New successes are marked with the help of increased computational power and AI becomes data-driven. In 1997, IBM’s DeepBlue beats world champion Kasparov at chess. In 2002, Amazon uses automated systems to provide product recommendations. In 2011, Apple releases Siri and IBM Watson beats two human champions at the TV quiz Jeopardy.

2012–today

Increased availability of data, connectedness and computational power allow for breakthroughs in machine learning, mainly in neural networks and deep learning, heralding a new era of increased funding and optimism about the AI potential. In 2012, Google driverless cars navigate autonomously and in 2016 Google AlphaGo beats a world champion in the complicated board game Go.

The Present and Future of AI

http://www.gov.cn/zhengce/content/2017-07/20/content_5211996.htm



中华人民共和国中央人民政府



(三) 战略目标。

分三步走：

第三步，到2030年人工智能理论、技术与应用总体达到世界领先水平，成为世界主要人工智能创新中心，智能经济、智能社会取得明显成效，为跻身创新型国家前列和经济强国奠定重要基础。

——形成较为成熟的新一代人工智能理论与技术体系。在类脑智能、自主智能、混合智能和群体智能等领域取得重大突破，在国际人工智能研究领域具有重要影响，占据人工智能科技制高点。

——人工智能产业竞争力达到国际领先水平。人工智能在生产生活、社会治理、国防建设各方面应用的广度深度极大拓展，形成涵盖核心技术、关键系统、支撑平台和智能应用的完备产业链和高端产业群，人工智能核心产业规模超过1万亿元，带动相关产业规模超过10万亿元。

——形成一批全球领先的人工智能科技创新和人才培养基地，建成更加完善的人工智能法律法规、伦理规范和政策体系。

The Present and Future of AI



(四) 主要目标

到2020年，基本完成适应新一代人工智能发展的高校科技创新体系和学科体系的优化布局，高校在新一代人工智能基础理论和关键技术研究等方面取得新突破，人才培养和科学研究的优势进一步提升，并推动人工智能技术广泛应用。

到2025年，高校在新一代人工智能领域科技创新能力和人才培养质量显著提升，取得一批具有国际重要影响的原创成果，部分理论研究、创新技术与应用示范达到世界领先水平，有效支撑我国产业升级、经济转型和智能社会建设。

到2030年，高校成为建设世界主要人工智能创新中心的核心力量和引领新一代人工智能发展的人才高地，为我国跻身创新型国家前列提供科技支撑和人才保障。

Course Topics

This course concentrates on general principles of rational agents and on components for constructing them.

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i> , in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i> , 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Course Topics

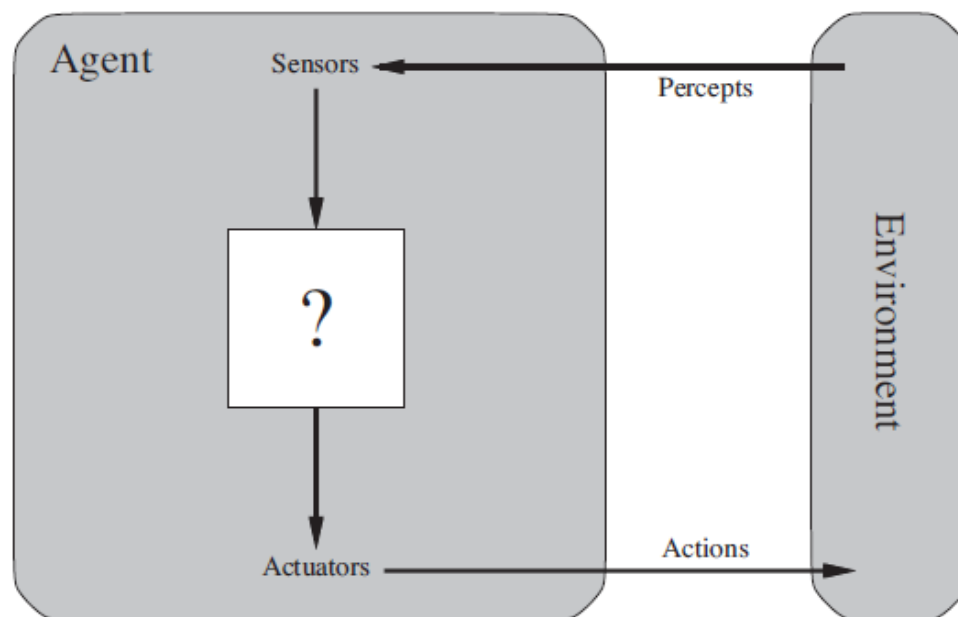
- Designing rational agents
 - An **agent** is an entity that perceives and acts
 - A **rational agent** selects actions that **maximize its expected utility**
 - Characteristics of the percepts, environment, and action space dictate techniques for selecting rational actions
- This course is about
 - General AI techniques for a variety of problems
 - Learning to recognize when and how a new problem can be solved with an existing technique

Course Topics

SES#	DATES		TOPICS
L1	Mar. 1		Introduction to AI
L2	Mar. 8	Problem -Solving	Search I: Uninformed Search + Heuristic Search
L3	Mar. 15		Search II: Adversarial Search + Monte Carlo Tree Search <i>Recitation I: Python Tutorial</i>
L4	Mar. 22		Constraint Satisfaction Problems
L5	Mar. 29	Logic	Logic + Inference
L6	Apr. 5	Uncertain Reasoning	Probability <i>Recitation II:</i>
L7	Apr. 12		Bayesian Network I: Representation
L8	Apr. 19		Bayesian Network II: Inference
L9	Apr. 26		Markov Network: Representation + Inference
L10	May 3		Lab <i>Recitation III:</i>
L11	May 10	Learning	Learning for PGM
L12	May 17		Classification
L13	May 24		Deep Learning
L14	May 31		Markov Decision Process
L15	Jun. 7		Reinforcement Learning <i>Recitation VI:</i>
L16	Jun. 21		<i>Final Project Presentation</i>

Intelligent Agents

- An **agent** is anything that can be viewed as **perceiving its environment** through **sensors** and **acting** upon that environment through **actuators**.
- For each possible **percept sequence**, a **rational agent** should select an action that is expected to **maximize its performance measure**, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.



Task environment

- Performance
- Environment
- Actuators
- Sensors

Agent = Architecture + Program

Task Environments

- Fully observable vs. partially observable
- Deterministic vs. stochastic
- Episodic vs. sequential
- Static vs. dynamic
- Discrete vs. continuous
- Known vs. unknown



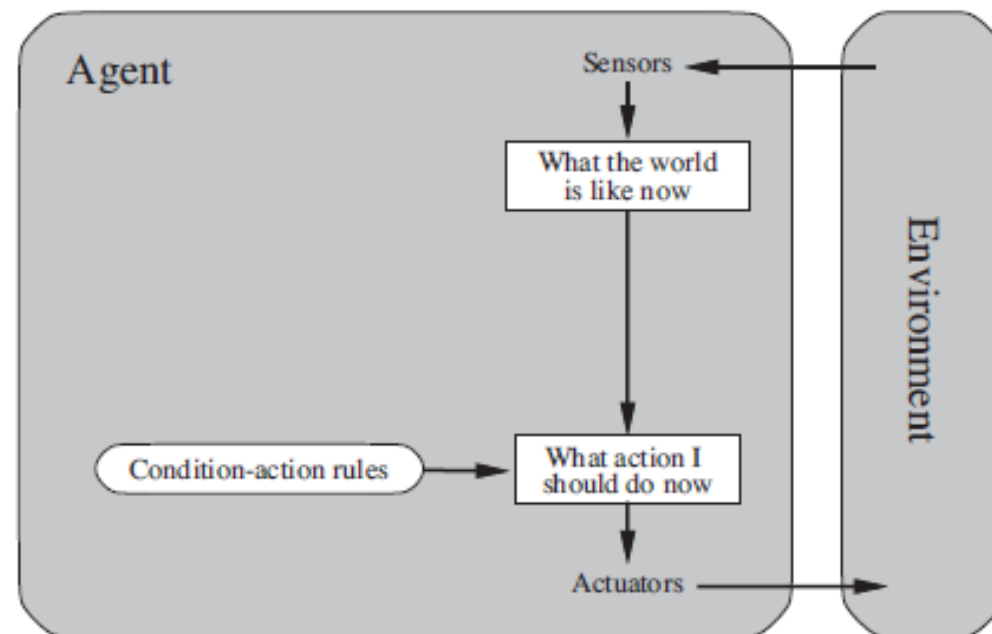
Task Environments

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

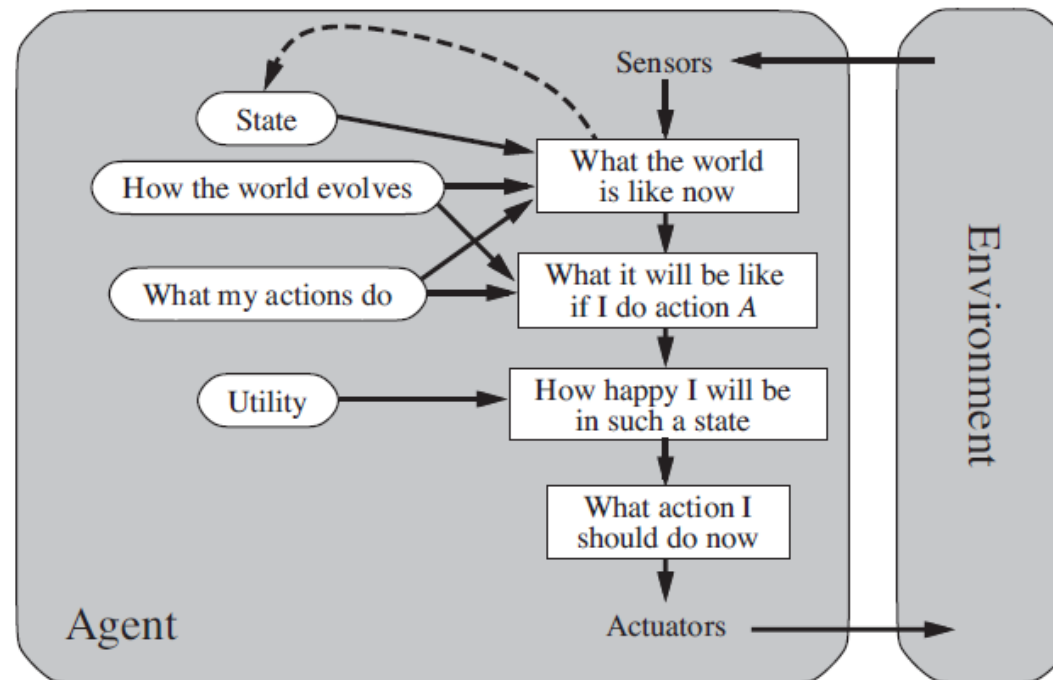
Reflex Agents

- Select action based on **current percept (and maybe memory)**
- May have memory or a model of the world's current state
- Do not consider the future consequences of their actions
- Consider how the world **IS**
- **Can a reflex agent be rational?**



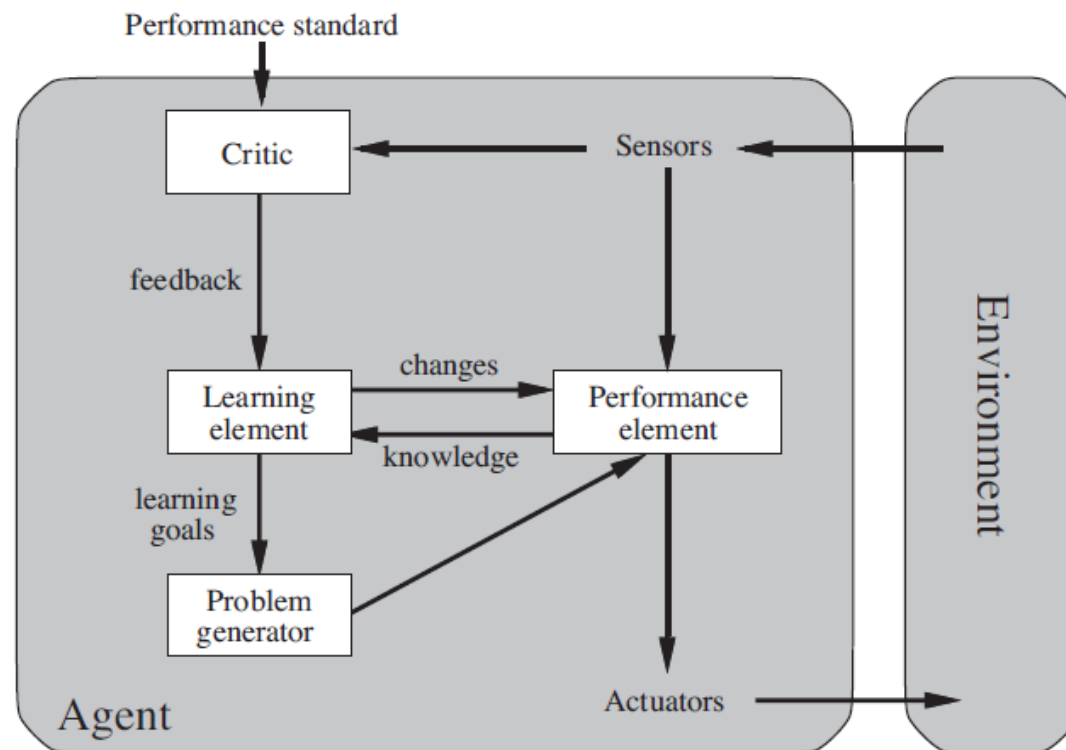
Planning Agents

- Decisions based on (hypothesized) consequences of actions
- Must have a model of how the world evolves in response to actions
- Must formulate a goal / utility
- Consider how the world **WOULD BE**



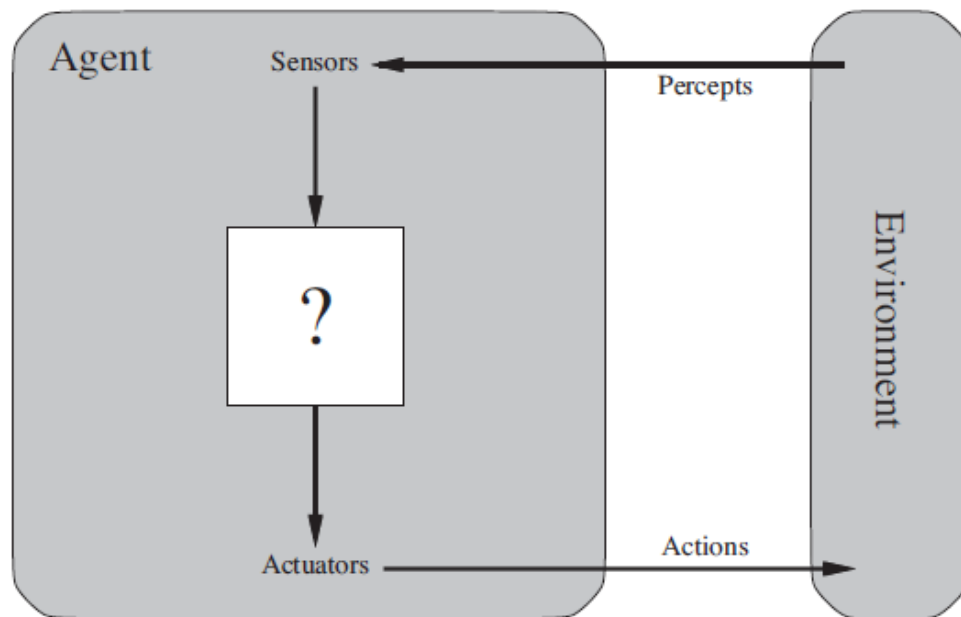
Learning Agents

- Learning element: making improvement
- Performance element: selecting external actions
- Problem generator: suggesting actions that will lead to new and informative experiences



Review

- Rational Agents
 - Structure



Task environment

- Performance
- Environment
- Actuators
- Sensors

Agent = Architecture + Program

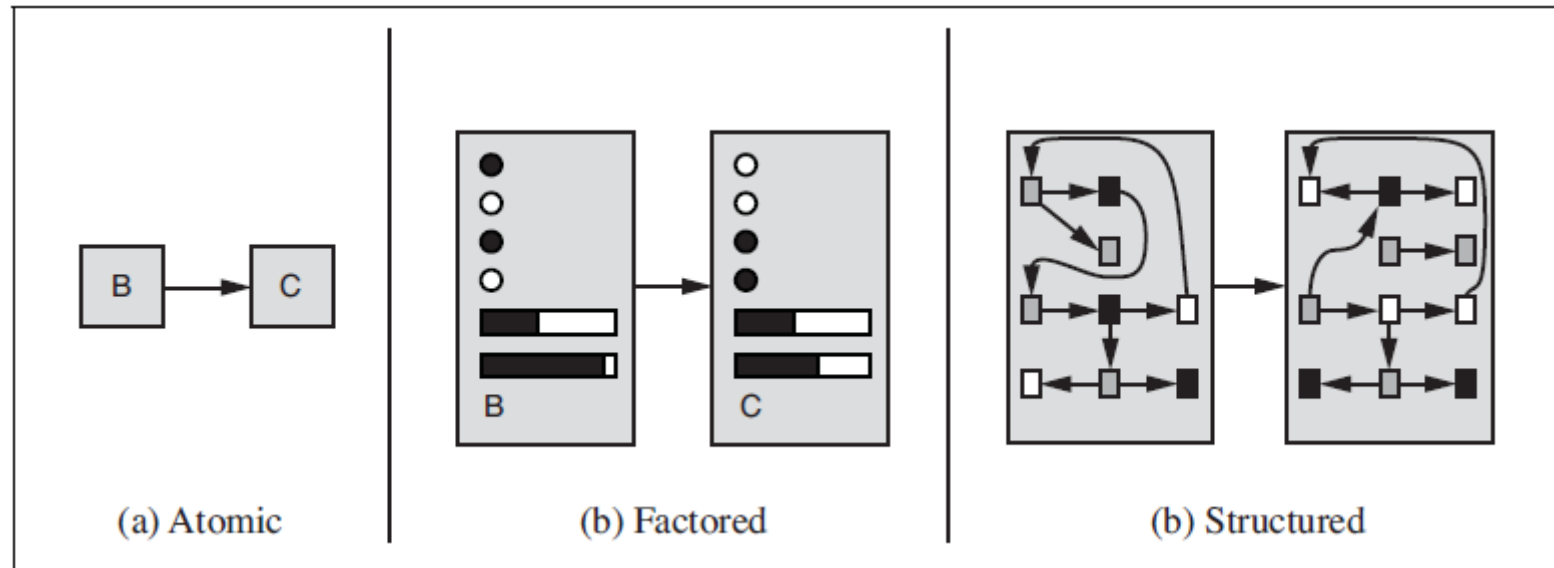
Review

- Rational Agents
 - Representation of Environments

Search
Game-playing
Markov decision processes

Constraint satisfaction
Bayesian network
Machine learning

First-order logic
Knowledge-based learning
Natural language processing



Assignments

- Reading assignment:
 - Ch. 1-2, Ch. 26-27